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Measurements of the gravitational constant using two independent methods

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Supplementary Information for “Measurement of the gravitational constant using two independent methods”

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Section 1: Experimental method

In this work, we have performed new determination of G using torsion pendulum with two completely independent methods, the time-of-swing (ToS) method and the angular acceleration feedback (AAF) method. The two methods, with a different set of systematic errors, have been carried out on independent apparatus, so that unknown systematic errors existed in one method would be unlikely to exist in the other. We chose these two methods because that the ToS method is the most frequently used in G measurement, while the AAF method has produced the most precise value of G up to date.

The ToS method:

The ToS method measures the change of the torsional oscillation frequency of a pendulum with the source masses at two different positions (named as “near” and “far” configuration, respectively). At the “near” position, the equilibrium position of the pendulum is in line with the source masses, and at the “far” position, it is perpendicular. The gravitational torque on the pendulum with a small angle θ from its equilibrium position may be expanded as $\tau_g(\theta) = -K_{1g}\theta - K_{3g}\theta^3 + O(\theta^5)$, where

$$K_{1g} = \frac{\partial^2 V_g(\theta)}{\partial \theta^2} \text{ and } K_{3g} = \frac{1}{6} \frac{\partial^4 V_g(\theta)}{\partial \theta^4} \text{ are evaluated at the “near” position or the “far”}$$

position, with $V_g(\theta)$ being the gravitational potential energy between them. Since K_{3g} and higher terms are neglected, K_{1g} is treated as the effective gravitational torsion constant and written as GC_g , where C_g is determined by the mass distributions of the pendulum and source masses. The frequency squared of the pendulum in the presence

of the source masses at each position is $\omega_{n,f}^2 = \frac{K_{n,f} + GC_{gn,gf}}{I}$, where I is the

pendulum’s moment of inertia about the fiber, $K_{n,f}$ is the torsion spring constant of the fiber and the subscripts n and f denote the “near” and “far” positions, respectively.

Then G can be determined by $G = \frac{I(\omega_n^2 - \omega_f^2) - (K_n - K_f)}{c_{gn} - c_{gf}} = \frac{I\Delta\omega^2}{\Delta C_g} \left[1 - \frac{\Delta K}{I\Delta\omega^2} \right]$, where ΔK denotes the correction of the fiber’s anelasticity. Considering the effect of the magnetic damper, the G value is given by¹⁹:

$$G = \frac{I\Delta\omega^2}{\Delta C_g} \left[1 - \frac{\Delta K}{I\Delta\omega^2} + \frac{I_m K_m^2}{IK_m^2} \right],$$

where I_m and K_m represent the moment of inertia of the magnetic damper and the torsion constant of the prehanger fiber, respectively (Supplementary Table 1). The values of $\Delta C_g/I$ in different measurements are shown in Supplementary Table 2.

This method has a considerable advantage of requiring neither a reference calibrated force nor precision measurement of angular displacements. However, this method is subject to systematic errors associated with the subtle property of the torsion fiber, especially the anelasticity. Since 1980s, our group has been measuring G with this method, and obtained G values in 1998 (HUST-99¹⁶), 2005 (HUST-05¹⁷), and 2009 (HUST-09^{18,19}). We have studied adequately almost all the fiber-dependent errors except for anelasticity. Indeed, the anelastic property of a torsion fiber is still under

investigation, and could be the largest uncertainty in HUST-09 experiment²³. Kuroda predicted that the anelasticity of the torsion fiber may lead to an upward correction of $1/(\pi Q)$ to the G value²¹, while Newman *et al.* suggested the correction should fall within a bound between 0 and $1/(2Q)$ ²², where Q is the quality factor of the torsional oscillation mode of the pendulum. To reduce such a systematic error, high Q fused silica fibers²⁰ are used in our current measurements using the ToS method on two independent apparatus.

The AAF method

In the AAF method, a thin planar pendulum is placed on a continuously rotating turntable (the pendulum turntable) located between two pairs of source masses rotated coaxially by another turntable (the source mass turntable). First, the pendulum turntable is rotated at a constant rate, so that the pendulum experiences a sinusoidal gravitational torque produced by the source masses. Then, a feedback loop turned on, the twist angle of the torsion balance is minimized to about zero by changing the angular velocity $\omega(t)$ of the pendulum turntable, and thus the angular acceleration of the pendulum is equal to the gravitational angular acceleration generated by the source masses. Meanwhile the angular velocity $\omega_a(t)$ of the source mass turntable is controlled to maintain a constant difference (ω_d) between the angular velocity of the two turntables by another feedback loop, so that $\omega_a(t) = \omega_d + \omega(t)$. Both $\omega(t)$ and $\omega_a(t)$ vary as a sinusoid with the same amplitude. When the two feedback loops work cooperatively well, the angular acceleration signal of the pendulum turntable of interest appears at $2\omega_d$. The relative angular position between the two turntables (or the pendulum and the source mass) changes with time as $\omega_d t$. This scheme is helpful to cleanly separate the signal from the lab-fixed gravitational background.

The gravitational angular acceleration on the torsion pendulum produced by the source masses can be expanded in spherical multipole moments as^{15,48} $\alpha(\phi) =$

$$-\frac{4\pi IG}{I} \sum_{l=0}^{\infty} \sum_{m=-l}^l \frac{1}{2l+1} m Q_{lm} q_{lm} e^{-im\phi}. \text{ Here } I \text{ is the pendulum moment of inertia, } \phi$$

is the relative azimuth angle between the two turntables, $q_{lm} = \int \rho(\vec{r}_t) r_t^l Y_{lm}^*(\theta_t, \phi_t) d^3 r_t$ are the multipole moments of the pendulum and $Q_{lm} = \int \rho(\vec{r}_a) r_a^{-l-1} Y_{lm}(\theta_a, \phi_a) d^3 r_a$ are the multipole fields of the external mass distribution, where the subscripts t and a denote the pendulum and the source masses, $\rho(\vec{r}_t)$ and $\rho(\vec{r}_a)$ denote their mass distribution densities, respectively. We define a

gravity coupling coefficient $P_{g,l,m} = \frac{8\pi}{I} \frac{1}{2l+1} m Q_{lm} q_{lm}$ ($m=1,2,\dots,l$). Since the

specially designed symmetric configuration of the pendulum and the source masses, the biggest term in $P_{g,l,m}$ is the one with $l=m=2$. Other small terms with l from 3 to 10, $m=2$ are all calculated and corrected in the results. When the two feedback loops work cooperatively well, the twist of the torsion fiber is then minimized to about zero, and the inertia torque equals to the gravitational torque. According to the discussion in ref. 24, the G measurement can be simplified as

$$G = \frac{1}{\left| \sum_{l=2}^{\infty} P_{g,l,2} \right|} \left(1 + \frac{K}{K_m} \frac{I_m}{I} \right) \alpha_t (2\omega_d)$$

where α_t is the angular acceleration amplitude of the torsion pendulum turntable, $\frac{K}{K_m} \frac{I_m}{I}$ is an additional correction arisen from the magnetic damper, I_m and K_m represent the moment of inertia of the magnetic damper and the torsion constant of the prehanger fiber, respectively, as listed in [Supplementary Table 1](#). K_m is obtained by measuring the period of the free oscillation when the prehanger fiber suspends the magnetic damper and a load (the mass equal to that of the G pendulum). K is calculated from the moment of inertia and the period of the pendulum. The $P_{g,l,2}$ in different measurements are shown in [Supplementary Table 3](#). Compared with the ToS method, as the torsion fiber does not experience any appreciable twist, the AAF method is insensitive to the torsion fiber properties, especially the anelasticity.

Section 2: Alignment and positioning

The position of the spheres relative to the pendulum is much less critical. In each method, the alignment between the pendulum and the source masses was performed with great care as follows:

The ToS method:

1. The ultra-low thermal expansion (ULE) glass disk is first mounted on the source mass turntable. Two spheres, supported by the three-point mounts²⁸, are fixed on the ULE disk. The center of the set of source masses is aligned with the rotation axis of the turntable with an uncertainty of no more than 5 μm guided with a coordinate measuring machine (CMM), which contribute less than 0.4 ppm uncertainty to the G value. Two additional three-point mounts that used as the gravitational counterbalances are positioned at the orthogonal locations symmetrically. The uncertainty of the 90° rotation angle of the turntable between the “near” and “far” positions is determined to be less than 134 μrad , which contributes less than 0.01 ppm to the G value.
2. The tilt of the ULE disk plane is adjusted by a level gauge. During the switching of the “near” and “far” positions, the deviation of the tilt is determined to be less than 20 μrad , which results in an uncertainty of 2 μm to the height of the source masses (less than 0.07 ppm).
3. With respect to the ULE disk plane, the centric heights of the source masses are measured by using the CMM and checked out by the bridge method¹⁹. The heights are determined with an uncertainty of less than 16 μm , which contributes less than 1.06 ppm uncertainty to the G value.
4. By using two glass gauges with a height difference of 10(1) μm , the centric height of the pendulum is adjusted to coincide with that of the source masses within 22 μm . The determined uncertainty is less than 17 μm , which contributes less than 0.47 ppm uncertainty to the G value.
5. With the help of an infrared detector, the rotation axis of the turntable is aligned with the torsion fiber in vacuum with uncertainty of no more than 19 μm that contributes a 1.25 ppm uncertainty to the G value.
6. By rotating the pendulum about the torsion fiber by 180°, the attitude of the pendulum about two orthogonal horizontal axes is measured to be 0.01~4 mrad with an uncertainty of less than 10 μrad by using an autocollimator, which contributes an uncertainty of less than 0.05 ppm to the G value.
7. The angle between the central line of the source masses and the equilibrium of the pendulum is adjusted to be less than 74 μrad by using a “transferring-reference method” as used in HUST-09¹⁹, and the result only contributes 0.04 ppm uncertainty to the G value.

The AAF method:

1. The ULE shelf is first mounted on the source mass turntable. The four spheres, supported with the three-point mounts²⁸, are mounted symmetrically centered on

the ULE shelf with upper and lower layers under CMM. The off-center displacement of the spheres from the rotation axis of the source mass turntable is adjusted to be less than 0.03 mm relative to their nominal positions, which brings in a 0.01 ppm uncertainty to the G value.

2. The tilts of the source mass turntable and the pendulum turntable are adjusted to be less than 89 and 40 μrad by inserting thin blocks under the baseplate of the turntables, which result less than 0.01 and 0.20 ppm uncertainties, respectively.
3. The torsion fiber is aligned with the rotation axis of the pendulum turntable within 0.45 mm by adjusting the fiber's suspension point, which corresponds to a 0.04 ppm uncertainty to the G value.
4. The alignment of the two turntable axes is adjusted to be less than 85 μm through moving the source mass turntable, which only contributes an uncertainty of 0.01 ppm to the G value.
5. A polished ULE gauge is used to detect the height difference of the pendulum and the set of source masses, and the result is less than 0.27 mm that contributes a 0.61 ppm uncertainty to the G value.
6. The attitude of the pendulum is measured to be 0.2~5 mrad with an uncertainty of less than 69 μrad by using an autocollimator, which contributes an uncertainty of less than 0.06 ppm to the G value.

Section 3: The nonlinear effects in the ToS method

Here we consider the nonlinear properties in which the period of the pendulum varies with its amplitude as^{49,50}: $T(A) = T_0 \left(1 + \frac{I\beta^2}{2K} - \frac{3K_3}{8K} A^2 \right)$, where K_3 is the nonlinear coefficient of the pendulum, A is the amplitude, β is the damping coefficient and T_0 is the unperturbed period, respectively. In the G measurement with the ToS method, K_3 contains three parts: the nonlinear effect of the fiber (K_{3f}), the nonlinear components of the gravitational field due to the source masses (K_{3g}) and the background gravity (K_{3b}), respectively. The nonlinear property of the silica fiber (K_{3f}) is studied by means of the method described in ref. 49 by using a quartz disk pendulum. The determined coefficient of K_{3f}/K is $-3(6) \times 10^{-3} \text{ rad}^{-2}$, which gives a null result. We choose $|K_{3f}/K| \leq 9 \times 10^{-3} \text{ rad}^{-2}$ to estimate the uncertainty to the G measurement.

The nonlinear coefficient of the gravitational field due to the source masses is related to the gravitational potential V_g between the pendulum and the source masses by $K_{3g} = \frac{1}{6} \frac{\partial^4 V_g(\theta)}{\partial \theta^4}$, where θ is the angle between them. Using the amplitude of the pendulum with an uncertainty of less than $56 \text{ } \mu\text{rad}$ and the misalignment between the pendulum and source masses, this nonlinear component is corrected synchronously in determining $\Delta\omega^2$. The resultant corrections for G value are shown in [Supplementary Table 4](#).

The lab-fixed background gravitational field is compensated with approximately 800 kg lead blocks with the method introduced in ref. 23. The nonlinear effect due to the background gravitational field (K_{3b}) is reduced to less than 0.14 ppm to the value of G .

Section 4: The data acquisition and correction factors

In the ToS method, all the data, include the pendulum twist, the temperature, seismic disturbances and fluctuations of the air pressure, are taken at a regular interval of 0.5 s triggered by a rubidium clock with a stability of 1×10^{-11} @1s and a frequency accuracy of $\leq 1 \times 10^{-10}$. The personal computers containing a 16 bits analog-to-digital converter are used to record the data.

In the AAF method, two extra 47,200 and 40,000 line/rev optical encoders are mounted on the rotors of the pendulum turntable and the source mass turntable, respectively. For each encoder, two read heads installed 180° apart are used to record the rotating angle signal. A data-acquisition system (DAS) with 16 bits analog-to-digital converters (ADCs), DACs, and 32 bits counters are used to record the data and control the experiment. The sampling rate of the DAS is 20 kHz, which is synchronized by a rubidium clock (the same type as that in the ToS method). The original data obtained in each first half second are averaged, then saved by the computer in the latter half exactly, so the sampling interval is 1 s. The angle of the pendulum turntable can be expressed as:

$$\theta = A \sin(2\omega_d t)$$

with A the amplitude of the angle and ω_d the angular velocity difference of the two turntables. The original data obtained in each first half second are averaged as:

$$\bar{\theta} = \frac{\int_{T-\Delta t/2}^{T+\Delta t/2} A \sin(2\omega_d t) dt}{\Delta t} = \frac{A \sin(\omega_d \Delta t)}{\omega_d \Delta t} \sin(2\omega_d T)$$

where $\Delta t=0.5$ s is the average time, $\omega_d=7.853982(3)$ mrad/s in AAF-I and $5.235988(4)$ mrad/s in AAF-II and AAF-III, respectively. Thus the attenuation of the true amplitude is $\frac{\sin(\omega_d \Delta t)}{\omega_d \Delta t}$, which introduces a correction of 2.57(1) ppm in AAF-I and 1.14(1) ppm in AAF-II and AAF-III, respectively.

In data processing, the pendulum turntable angle is numerically differentiated once with a time increment of ΔT and yields an angular velocity:

$$\begin{aligned} \omega^* &= \frac{A \sin(2\omega_d(t + \Delta T/2)) - A \sin(2\omega_d(t - \Delta T/2))}{\Delta T} \\ &= \frac{2A \sin(\omega_d \Delta T)}{\Delta T} \sin\left(2\omega_d t + \frac{\pi}{2}\right) \end{aligned}$$

Compared with the real angular velocity of $\omega = 2A\omega_d \sin\left(2\omega_d t + \frac{\pi}{2}\right)$, an attenuation coefficient of $\frac{\sin(\omega_d \Delta T)}{\omega_d \Delta T}$ is introduced. Since the pendulum turntable angle is numerically differentiated twice with a time increment of $\Delta T=10$ s to yield the angular acceleration, the true amplitude of the angular acceleration is attenuated by

$\left(\frac{\sin(\omega_d \Delta T)}{\omega_d \Delta T}\right)^2$, which introduces a correction of 2,058.71(1) ppm in AAF-I and 914.35(1) ppm in AAF-II and AAF-III, respectively. Different time increment are used for the calculation and we find that there are some influence on the signal at high frequency, but the interested signal is independent of the choice of ΔT after corrected by the actuation coefficient.

Section 5: Compensation of the background gravitational gradient in the AAF method

The co-moving background gravitational gradient:

In the AAF method, the mass asymmetry and imperfection of the ULE shelf and the rotating parts of the source mass turntable can generate a gravitational signal on the pendulum at the interested signal frequency ($2\omega_d$). The co-moving background gravity gradient effect is measured with a free torsion pendulum with the spheres removed. The source mass turntable is rotated at 0.5 mHz, but the pendulum turntable keeps static. The interested signal appeared at 1 mHz is found to be 1,195(9) ppm. After compensated by a set of specially fabricated copper and aluminium blocks adhered on the shelf (Supplementary Fig. 1), this effect is reduced to only 1.86 ppm, 1.35 ppm, and 1.72 ppm to the value of G in three individual experiments, respectively. Furthermore, by using the CMM, the possible deformation of the upper layer shelf after placing the spheres is measured to be less than 8 μm , which contributes an uncertainty of 1.51 ppm to the value of G .

The lab-fixed background gravitational gradient:

One of the advantages of the AAF method is that it can remove the lab-fixed gravitational gradient interaction in frequency domain. This advantage allows us to perform the G measurement under a non-zero lab-fixed background gravitational gradient field. Therefore, we only used 8 blocks of lead (with each mass of ≈ 34 kg) to make a preliminary compensation for the lab-fixed gravitational gradient field, rather than completely compensating it painstakingly (Supplementary Fig. 2). After compensation, the amplitude of residual background signal at $2\omega_l$ is about 77 nrad/s^2 , while the signal of interest from the source masses is at $2\omega_d$ with an amplitude of about 462 nrad/s^2 , which result the superimposed signal not a perfect sine wave as shown in the time-domain (Fig. 2e). The cost of not perfectly compensating the lab-fixed gravitational gradient is that the choose of the angular velocity of the two turntables are not arbitrary, it should let the signal frequency of interest ($2\omega_d$) away from that of the lab-fixed background and its harmonics.

Section 6: The combination of the values of G

In the ToS method, the measurement of G with fibers 1, 3 and 4 have all been performed twice. For each fiber, we obtain two values of G , which are denoted as $G_{k,a}$ and $G_{k,b}$ ($k=1, 3, 4$) with uncertainties of $u_{k,a}$ and $u_{k,b}$, respectively. Their relative weights are calculated as $p_{k,a} = \frac{u_{k,a}^{-2}}{u_{k,a}^{-2} + u_{k,b}^{-2}}$ and $p_{k,b} = \frac{u_{k,b}^{-2}}{u_{k,a}^{-2} + u_{k,b}^{-2}}$, respectively. Then the weighted mean value of G for fiber k is

$$G_k = p_{k,a}G_{k,a} + p_{k,b}G_{k,b}.$$

With an uncertainty of

$$u_k = \sqrt{p_{k,a}^2 \sum_{i=1}^n u_{k,a,i}^2 + p_{k,b}^2 \sum_{j=1}^n u_{k,b,j}^2 + 2p_{k,a}p_{k,b} \sum_{1 \leq i \leq j} \gamma_{k,i,j} u_{k,a,i} u_{k,b,j}}$$

where $u_{k,a,i}$ ($u_{k,b,j}$) is the uncertainty of the i th (j th) error item of the experimental run a (b) with the fiber k , and $\gamma_{k,i,j}$ is the correlation coefficient between them, n is the total number of error items. For the two experiments with the same fiber, the change of frequency squared $\Delta\omega_{back}^2$ with the absence of the source masses (background effect) is measured once with an uncertainty of $u_{k,back}$. The change of frequency squared $\Delta\omega_{nf}^2$ with the source masses being present is measured twice with uncertainties of $u_{k,a,\omega}$ and $u_{k,b,\omega}$, which are mainly limited by the random thermal noise and statistically independent. For the same fiber, the correlation between the same error item of the two experimental runs except for $\Delta\omega^2$ is considered to be 100% as the same apparatus is used. Therefore, we obtain the combined uncertainty of the G value for one fiber as

$$u_k = \sqrt{\sum_{i=1}^m (p_{k,a}u_{k,a,i} + p_{k,b}u_{k,b,i})^2 + u_{k,back}^2 + p_{k,a}^2 u_{k,a,\omega}^2 + p_{k,b}^2 u_{k,b,\omega}^2}$$

where $m=n-1$ is the number of error items except for $\Delta\omega^2$. Since the measurement of G with fiber 2 has been performed only once, its result is directly noted as G_2 with an uncertainty of

$$u_2 = \sqrt{\sum_{i=1}^m u_{2,i}^2 + u_{2,\omega}^2}$$

where $u_{2,i}$ is the i th uncertainty item except for $\Delta\omega^2$ in the measurement of G with fiber 2, and $u_{2,\omega}$ is the statistical uncertainty of $\Delta\omega^2$. With the method as discussed above, we determine four values of G for fibers 1-4, which are denoted as G_k with uncertainties of u_k , respectively, where $k=1, 2, 3$ and 4. The relative weight for fiber k is calculated as $p_k = \frac{u_k^{-2}}{u_1^{-2} + u_2^{-2} + u_3^{-2} + u_4^{-2}}$. Therefore, the weighted mean value of G determined by the four fibers in the ToS method is

$$G_{ToS} = \sum_{k=1}^4 p_k G_k$$

The correlation between different fibers is estimated as follows. The uncertainties of $\Delta\omega^2$ are mainly limited by the random thermal noise and statistically independent between different fibers, thus the correlation coefficient is 0. Though the two apparatus (including the pendulum and the source masses) are completely independent in ToS-I and ToS-II, we conservatively assume that there is a 100% correlation of the same error items except for the statistical $\Delta\omega^2$ between the four results considering the common measurement method used for some key parameters. Thus the combined uncertainty of G is

$$u_{ToS} = \sqrt{\sum_{i=1}^m (p_1 u_{1,i} + p_2 u_{2,i} + p_3 u_{3,i} + p_4 u_{4,i})^2 + p_1^2 u_{1,\omega}^2 + p_2^2 u_{2,\omega}^2 + p_3^2 u_{3,\omega}^2 + p_4^2 u_{4,\omega}^2}$$

where $u_{k,i} = p_{k,a} u_{k,a,i} + p_{k,b} u_{k,b,i}$ ($k=1, 3, 4$) are the combined uncertainties for fiber k that contributed by the i th error item, $u_{k,\omega} =$

$\sqrt{u_{k,back}^2 + p_{k,a}^2 u_{k,a,\omega}^2 + p_{k,b}^2 u_{k,b,\omega}^2}$ ($k=1, 3, 4$) are the combined uncertainties of $\Delta\omega^2$ for fiber k , respectively, $u_{2,i}$ and $u_{2,\omega}$ for fiber 2 are defined as before.

In the AAF method, the three experimental runs give three values of G , which are denoted as G_a , G_b , and G_c with uncertainties of u_a , u_b , and u_c , respectively. The statistical error of the angular acceleration α_t is mainly limited by the random thermal noise and independent between the three experiments. Since the main parts of the apparatus including the pendulum and the source masses are the same, we think there is a 100% correlation between the three results of the same error item except for the statistical angular acceleration (α_t) of the torsion pendulum turntable in three experiments. With the same method as discussed above, we obtain the weighted mean value G_{AAF} with a combined uncertainty of u_{AAF} in the AAF method.

Section 7: Discussion of the results

The G values obtained by the two completely independent methods have the smallest uncertainty to date and both agree with the CODATA-2014 value within 2σ range. The value obtained in current work with the ToS method is a hundred ppm larger than our previous measurement (HUST-09). We have no definite explanation at present for the inconsistency of the two results. We suspect that the difference may come from some kind of potential systematic error in our previous work, such as the anelasticity of the lower- Q tungsten fiber.

In HUST-09 experiment, the anelastic effect was not corrected using the $1/(\pi Q)$ hypothesis²¹. Instead, the anelasticity contribution to the G value was directly measured to be (211.80 ± 18.69) ppm with two additional disk pendulums using a high- Q silica fiber ($Q \approx 3.36 \times 10^5$)²³, slightly greater than the Kuroda's hypothesis. In this measurement, in order to amplify the anelastic effect, the change of period is amplified to 93 s (the oscillation periods of two pendulums are 488.2 s and 581.2 s, respectively), about 30 times of the period change 3.2 s in the G measurement (the periods are 532.6 s and 535.8 s, respectively). The correction due to the fiber anelasticity in using the time-of-swing method can be expressed as^{18,23}

$$\frac{\Delta G}{G} = \frac{\Delta k}{I_G \Delta(\omega_{nf}^2)} = \frac{k(\omega_n) - k(\omega_f)}{I_G(\omega_n^2 - \omega_f^2)} = \frac{k(\omega_1) - k(\omega_2)}{I_G(\omega_1^2 - \omega_2^2)}$$

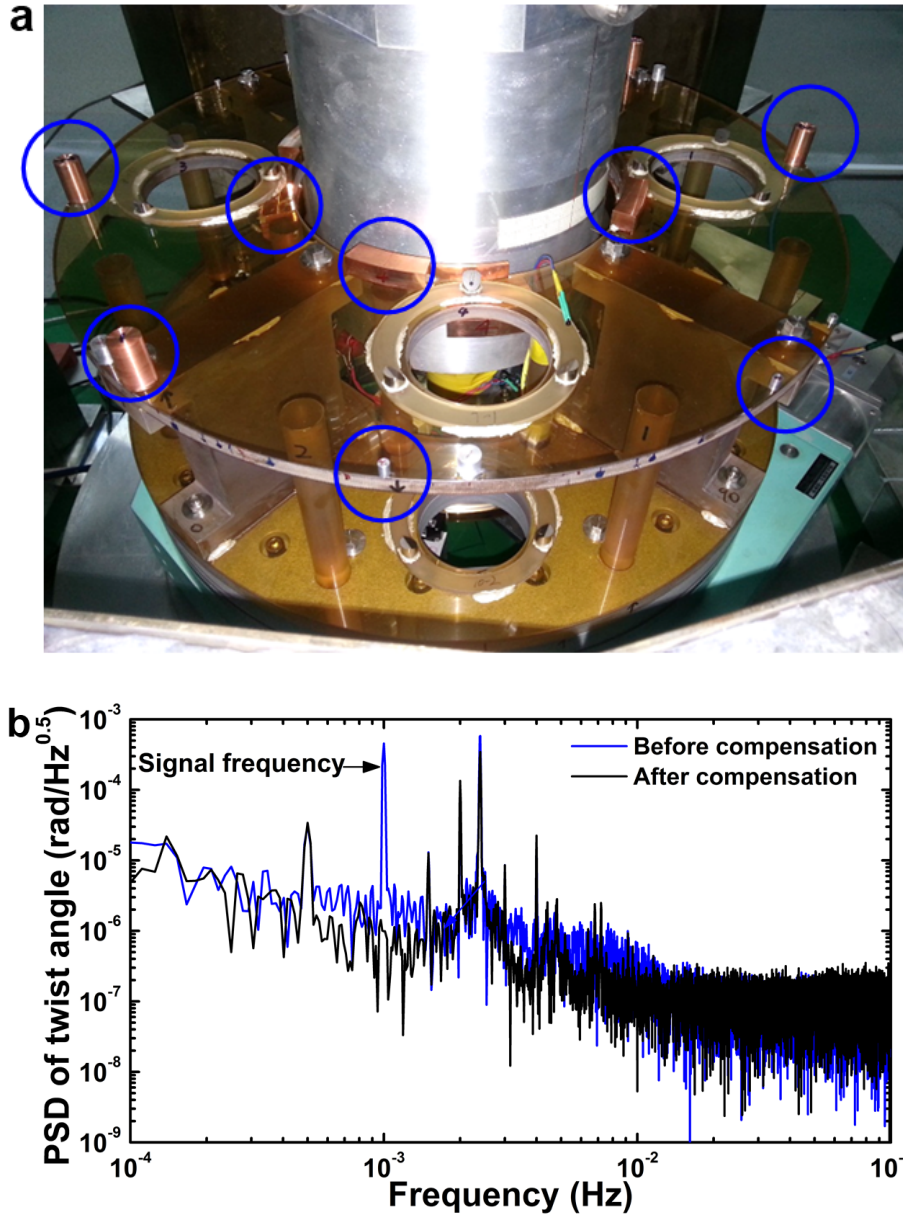
where $\omega_{n,f}^2$ is the frequency squared of the pendulum in the presence of the source masses at “near” and “far” positions in the G measurement, $\omega_{1,2}^2$ is the frequency squared of the additional pendulum used in the anelasticity measurement, $k(\omega)$ is the torsion spring constant of the fiber at the frequency ω , and I_G is the moment of inertia of the pendulum used in the G measurement. We have directly measured the anelasticity with a higher precision, but the experiment is based on a linear assumption that $\frac{k(\omega_n) - k(\omega_f)}{(\omega_n^2 - \omega_f^2)} = \frac{k(\omega_1) - k(\omega_2)}{(\omega_1^2 - \omega_2^2)}$ in the frequency interval between ω_1^2 and ω_2^2 . Although the assumption sounds reasonable due to the very small frequency range involved in the experiment, it required much more test at different frequency. In addition, as discussed by Desalvo, the anelastic effect of a fiber is intrinsically stochastic⁵¹ according to the Self Organized Criticality statistics⁵² of dislocations inside the metal grains. If this conjecture is correct, whether the state (different amplitudes, frequency, external excitation, and some other factors) of the tungsten fiber in the measurement of the anelasticity or the G measurement was changed, the measured anelasticity by additional experiment may be different from the value that in the measurement of G . So a possible solution for the time-of-swing method to achieve a small anelastic effect is using elastic material with less or no dislocations, such as the fused silica.

In our new measurement, coated silica fibers with Q of about 5×10^4 were used and the anelastic effect was minimized to several ppm. In addition, different silica fibers and

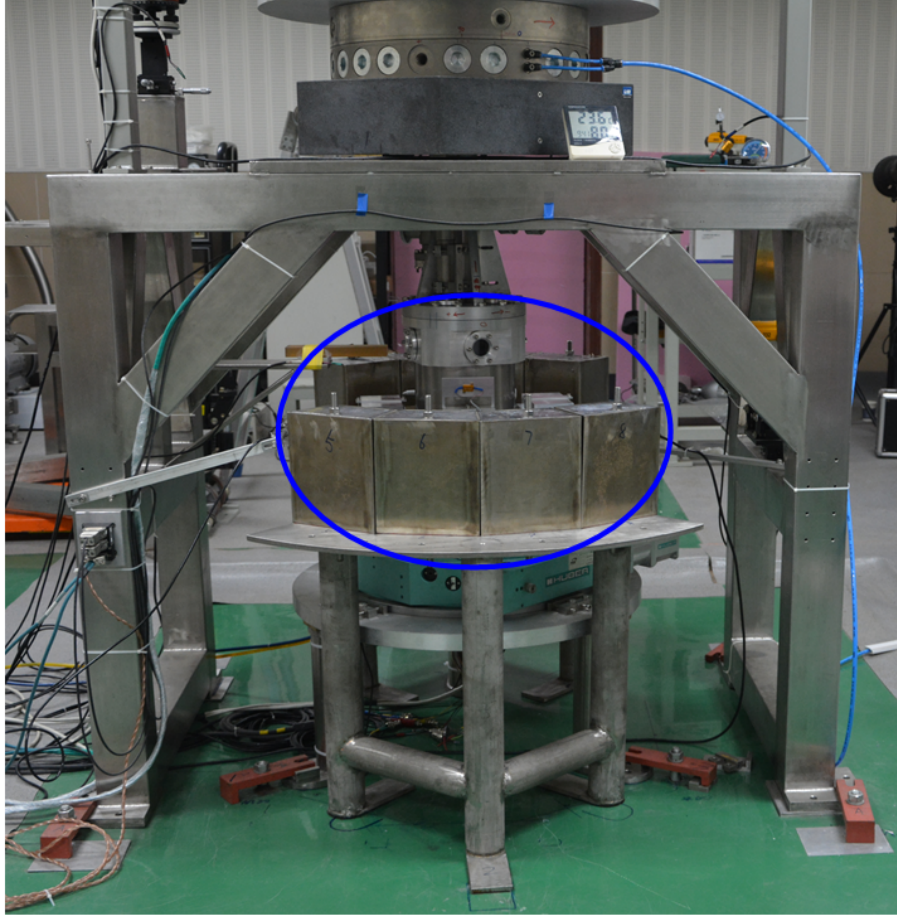
two independent apparatus with different pendulum and source masses are used to find the potential errors related with the fiber or the apparatus. The obtained results show good consistency with each other within the uncertainties. Moreover, we have conducted a series of improvements to reduce other large corrections and error terms existed in our previous experiment (see [Extended Data Table 3](#)). Meanwhile, the completely independent measurement of G with another method, the AAF method, were performed by other members in our group at the same period of time and the obtained results are consistent with the ToS method within 3σ range. After considering all these factors, we think that the new measurements have a higher level of confidence.

Supplementary References

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Supplementary Figure 1 | The PSD of the pendulum twist angle in AAF-I and the photo of the ULE shelf. The source mass turntable rotates at a frequency of 0.5 mHz, and the pendulum oscillates freely. The signal of interest occurs at 1 mHz. Before and after compensation, the amplitudes of the twist angle at the signal frequency are 2.946(21) and 0.002(2) μrad , respectively. After compensated by a set of specially fabricated copper and aluminium blocks adhered on the shelf (shown in the circles), this effect is reduced to less than 2 ppm to the value of G .



Supplementary Figure 2 | The photo of the lead blocks used to compensate the lab-fixed background gradient field. 8 blocks of lead with each mass of ≈ 34 kg (shown in the circles) are placed symmetrically along the East-West direction. After compensation, the amplitude of residual background signal at $2\omega_l$ is about 77 nrad/s^2 .

Supplementary Table 1 | The parameters of the two-stage pendulum system. For fiber 1-3 in the ToS method, the parameters are all converted to the values at 20.2 °C, and for fiber 4, the normal temperature during experimental runs is 21.5 °C. In the AAF method, the parameters are all converted to the values at 23.7 °C. Uncertainties are quoted as one standard deviation.

Items	$I_m(\times 10^{-5}$ kg·m ²)	$K_m(\times 10^{-5}$ Nm·rad ⁻¹)	$I(\times 10^{-5}$ kg·m ²)	$K(\times 10^{-9}$ Nm·rad ⁻¹)	$\Delta G/G$ (ppm)
ToS-I: Fiber 1	2.18(1)	1.2(1)	4.7705(1)	12.2(1)	0.47(8)
ToS-I: Fiber 2	2.18(1)	1.2(1)	4.7706(1)	47.4(1)	7.13(1.19)
ToS-I: Fiber 3	2.18(1)	1.2(1)	4.7705(1)	10.1(1)	0.32(5)
ToS-II: Fiber 4	2.21(2)	1.4(2)	4.6477(1)	10.6(1)	0.27(8)
AAF-I and II	2.401(2)	1.199(4)	2.776(1)	6.313(16)	455.40(1.95)
AAF-III	2.404(2)	21.24(3)	2.776(1)	6.313(16)	25.74(8)

Supplementary Table 2 | The determined $\Delta C_g/I$ and the average value of $\langle \Delta \omega^2 \rangle$ in the ToS method. For fibers 1, 3 and 4, the measurements were performed twice, and for fiber 2, the measurement was performed once. Uncertainties are quoted as one standard deviation.

Parameters		$\Delta C_g/I$ (kg·m ⁻³)	$\langle \Delta \omega^2 \rangle$ (×10 ⁻⁶ s ⁻²)	G (×10 ⁻¹¹ m ³ kg ⁻¹ s ⁻²)
ToS-I: Fiber 1	First experiment	24912.86(23)	1.662732(17)	6.674154(95)
	Repeated experiment	24912.12(23)	1.662699(18)	6.674222(98)
ToS-I: Fiber 2		24912.15(23)	1.662698(51)	6.674237(219)
ToS-I: Fiber 3	First experiment	24911.70(22)	1.662684(20)	6.674274(103)
	Repeated experiment	24911.72(22)	1.662683(17)	6.674266(94)
ToS-II: Fiber 4	First experiment	25003.05(25)	1.668719(23)	6.674017(117)
	Repeated experiment	25002.95(25)	1.668734(23)	6.674105(116)
Combined				6.674184(78)

Supplementary Table 3 | The determined $P_{g,l,m}$ and the average value of $\langle\alpha_t\rangle$ in the AAF method. l ranges from 2 to 10, $m=2$. Except for $P_{g,2,2}$, the corrections due to other terms amount to about 232 ppm. The $l>10$, $m=2$ terms in $P_{g,l,m}$ are negligible. α_t ($2\omega_d$) has been corrected for the air density effect. Uncertainties are quoted as one standard deviation.

Parameters	AAF-I	AAF-II	AAF-III
Average temperature ($^{\circ}\text{C}$)	22.8	23.7	23.7
$P_{g,2,2}$ ($\text{kg}\cdot\text{m}^{-3}$)	6924.7424-0.0517i	6924.7243-0.0044i	6924.8056+0.0087i
$P_{g,3,2}$ ($\text{kg}\cdot\text{m}^{-3}$)	-0.0046	-0.0048	-0.0048
$P_{g,4,2}$ ($\text{kg}\cdot\text{m}^{-3}$)	0.0001	0	0
$P_{g,5,2}$ ($\text{kg}\cdot\text{m}^{-3}$)	0	0	0
$P_{g,6,2}$ ($\text{kg}\cdot\text{m}^{-3}$)	1.5323+0.0001i	1.5330	1.5329
$P_{g,7,2}$ ($\text{kg}\cdot\text{m}^{-3}$)	0	0	0
$P_{g,8,2}$ ($\text{kg}\cdot\text{m}^{-3}$)	0.0814	0.0815	0.0815
$P_{g,9,2}$ ($\text{kg}\cdot\text{m}^{-3}$)	0	0	0
$P_{g,10,2}$ ($\text{kg}\cdot\text{m}^{-3}$)	0.0001	0.0001	0.0001
$ \sum_{l=2}^{10} P_{g,l,2} $ ($\text{kg}\cdot\text{m}^{-3}$)	6926.352(74)	6926.334(75)	6926.415(74)
$\langle\alpha_t(2\omega_d)\rangle$ (nrad/s ²)	462.0912(16)	462.0791(12)	462.2941(6)
G ($\times 10^{-11} \text{ m}^3\text{kg}^{-1}\text{s}^{-2}$)	6.674534(83)	6.674375(82)	6.674535(75)
Combined G ($\times 10^{-11} \text{ m}^3\text{kg}^{-1}\text{s}^{-2}$)	6.674484(78)		

Supplementary Table 4 | The nonlinearity effects of the gravitational field and the fiber in the ToS method with units of ppm. Uncertainties are quoted as one standard deviation.

Items		Gravitational nonlinearity (source masses)	Fiber nonlinearity
ToS-I: Fiber 1	First experiment	19.72(0.62)	0(1.45)
	Repeated experiment	22.14(0.62)	0(1.45)
ToS-I: Fiber 2		13.94(0.62)	0(4.84)
ToS-I: Fiber 3	First experiment	15.96(0.62)	0(1.10)
	Repeated experiment	14.02(0.62)	0(1.03)
ToS-II: Fiber 4	First experiment	17.90(0.22)	0(1.67)
	Repeated experiment	18.01(0.22)	0(1.26)